Graph Theory

Week 1 — Lecture 2

Graph Operations

Today's Plan

Graphical sequences & Havel–Hakimi

Finish key isomorphism ideas

Complement of a graph

Union, intersection & ring sum

Cartesian product

Direct/tensor product

Conceptual consolidation

Complement of a Graph

The complement $\bar{G}$ has the same vertices as G.

Two distinct vertices are adjacent in $\bar{G}$ exactly when they are not adjacent in G.

$$|E(G)| + |E(\bar{G})| = n(n-1)/2$$

$$\deg_{\bar{G}}(v) = n-1-\deg_G(v)$$

Useful result: if G is disconnected, then $\bar{G}$ is connected.

Three Set-Based Graph Operations

Keep this part simple

Union: $E(G_1 \cup G_2) = E_1 \cup E_2$

Intersection: $E(G_1 \cap G_2) = E_1 \cap E_2$

Ring sum: $E(G_1 \oplus G_2) = (E_1 \cup E_2) - (E_1 \cap E_2)$

Think of them as: ALL edges / COMMON edges / DIFFERENT edges.

Use one pair of small graphs to demonstrate all three operations.

Cartesian Product

The graph product to emphasize

Vertices of G □ H are ordered pairs (g,h).

Two vertices are adjacent when:

g is unchanged and h moves along an edge of H, OR

h is unchanged and g moves along an edge of G.

$$|V(G \square H)| = |V(G)| \times |V(H)|$$

Example: $P_2 \square P_3$

P_2 has 2 vertices and P_3 has 3 vertices.

Therefore, $P_2 \square P_3$ has 6 vertices.

Visual interpretation:

take one copy of P_3 for each vertex of P_2;

connect corresponding vertices according to the edge in P_2.

The result is a 2 × 3 grid.

Main idea: graph products construct larger structured networks from smaller graphs.

A Useful Extension: Hypercube

$$Q_n = K_2 \square K_2 \square \cdots \square K_2$$

Q_1: one edge

Q_2: a 4-cycle

Q_3: cube graph

Q_n has 2^n vertices and is n-regular.

Application idea: structured interconnection networks / state spaces.

Direct / Tensor Product

Definition only — no detailed exercises

G × H also uses ordered pairs (g,h).

Difference from Cartesian product:

- Cartesian: one coordinate moves while the other stays fixed.
- Direct/tensor: both coordinates must move along edges simultaneously.

Main takeaway: different adjacency rules produce different product graphs.

Conceptual Consolidation

1. Can an even degree sum guarantee that a sequence is graphical?
2. Can two non-isomorphic graphs have the same degree sequence?
3. If G is disconnected, what can we say about $\bar{G}$?
4. What is the conceptual difference between union, intersection and ring sum?
5. Which product naturally creates grid-like graphs?
6. In a Cartesian product, how many coordinates change along one edge?

Lecture Summary

Complement reverses adjacency and has simple degree/edge relationships.

Union, intersection and ring sum correspond to set operations on edges.

Cartesian product is the main graph product to retain; it builds structured networks.

Direct/tensor product is introduced only to distinguish its adjacency rule.

Next: Advanced Connectivity — cut vertices, bridges, vertex/edge connectivity and Menger's theorem.